Chapter 6. Multiple Linear Regression

Matrix form, least squares, ANOVA, expected mean squares, distribution theory, partial regression

Contents

6.1 The model in matrix form **2**

6.2 Least squares **2**

6.3 Analysis of variance **3**

6.4 Properties of the estimators **4**

6.5 Distribution theory **4**

6.6 Partial regression: the Frisch–Waugh–Lovell theorem **5**

6.7 Selected exercises **5**

6.1 The model in matrix form

With p explanatory variables,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \qquad i = 1, \ldots, n,$$

or, stacking the observations,

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n),$$

where $\mathbf{y}$ is $n \times 1$, X is the $n \times (p+1)$ **design matrix** whose i-th row is $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$, and $\beta = (\beta_0, \ldots, \beta_p)'$. We assume $n > p+1$ and $\operatorname{rank} X = p+1$. Simple regression is the case $p = 1$, with rows $(1, x_i)$.

6.2 Least squares

Motivation. Matrix notation unifies the formulas: $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$, a projection onto the column space of X.

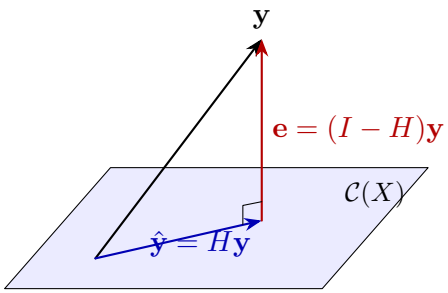


Figure. $\hat{\mathbf{y}}$ *is the orthogonal projection of* $\mathbf{y}$ *onto* $\mathcal{C}(X)$; $\mathbf{e} \perp \mathcal{C}(X)$ *is the normal equations* $X^\top \mathbf{e} = \mathbf{0}$.

Theorem 6.1

The least squares estimator minimizing $S(\beta) = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta)$ is the solution of the normal equations $X'X\hat{\beta} = X'\mathbf{y}$:

$$\hat{\beta} = (X'X)^{-1} X'\mathbf{y}.$$

Proof. $S(\beta) = \mathbf{y}'\mathbf{y} - 2\beta' X'\mathbf{y} + \beta' X'X\beta$ has gradient $-2X'\mathbf{y} + 2X'X\beta$ and Hessian $2X'X$, positive definite by full rank ($\mathbf{a}'X'X\mathbf{a} = \|X\mathbf{a}\|^2 > 0$), so the stationary point is the global minimum. Alternatively, for any β,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat{\beta}\|^2 + \|X(\hat{\beta} - \beta)\|^2,$$

because the cross term $2(\hat{\beta} - \beta)' X'(\mathbf{y} - X\hat{\beta})$ vanishes by the normal equations. ∎

For $p = 1$,

$$X'X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{n S_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix},$$

which reproduces the formulas of Chapter 3.

Example 6.1

For $n = 6$ observations of (x_1, x_2, y):

x_1	1	2	3	4	5	6
x_2	2	1	4	3	6	5
y	3	6	7	10	12	13

Centring gives $\bar{x}_1 = \bar{x}_2 = 3.5$, $\bar{y} = 8.5$, and the centred cross products $S_{11} = S_{22} = 17.5$, $S_{12} = 14.5$, $S_{1y} = 35.5$, $S_{2y} = 28.5$. The slopes solve

$$\begin{pmatrix} 17.5 & 14.5 \\ 14.5 & 17.5 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} 35.5 \\ 28.5 \end{pmatrix},$$

so, with determinant $17.5^2 - 14.5^2 = 96$,

$$\hat{\beta}_1 = \frac{17.5 \cdot 35.5 - 14.5 \cdot 28.5}{96} = 2.167, \qquad \hat{\beta}_2 = \frac{17.5 \cdot 28.5 - 14.5 \cdot 35.5}{96} = -0.167,$$

and $\hat{\beta}_0 = 8.5 - 3.5(2.167 - 0.167) = 1.5$. Although $S_{2y} > 0$, the coefficient of x_2 is slightly negative given x_1: x_2 adds little beyond x_1 (Section 6.6).

Fitted values $\hat{\mathbf{y}} = X\hat{\beta} = H\mathbf{y}$, **hat matrix** $H = X(X'X)^{-1}X'$, residuals $\mathbf{e} = (I - H)\mathbf{y}$. H is symmetric idempotent, $HX = X$, $\mathrm{tr}H = p + 1$: the projection onto $\mathrm{col}(X)$. So $X'\mathbf{e} = \mathbf{0}$: residuals sum to zero and are orthogonal to every regressor and to $\hat{\mathbf{y}}$.

6.3 Analysis of variance

With $J_n = \mathbf{1}\mathbf{1}'$, $n\bar{y}^2 = \mathbf{y}'(J_n/n)\mathbf{y}$ is the **correction term**. The sums of squares are quadratic forms:

$$SST = \mathbf{y}'\Big(I - \frac{J_n}{n}\Big)\mathbf{y},$$

$$SSE = \mathbf{y}'(I - H)\mathbf{y} = \mathbf{y}'\mathbf{y} - \hat{\beta}'X'\mathbf{y}, \qquad SSR = \mathbf{y}'\Big(H - \frac{J_n}{n}\Big)\mathbf{y} = \hat{\beta}'X'\mathbf{y} - n\bar{y}^2,$$

with $SST = SSR + SSE$. The degrees of freedom are the ranks of the matrices: $n - 1$, p and $n - p - 1$. (SST has $n - 1$ degrees of freedom because the n deviations $y_i - \bar{y}$ satisfy one linear constraint.)

source	SS	df	MS	F_0
regression	SSR	p	$MSR = SSR/p$	MSR/MSE
residual	SSE	$n - p - 1$	$MSE = SSE/(n - p - 1)$	
total	SST	$n - 1$		

Theorem 6.2 (Expected mean squares)

$$E(MSE) = \sigma^2, \qquad E(MSR) = \sigma^2 + \frac{1}{p}\beta'X'\Big(I - \frac{J_n}{n}\Big)X\beta.$$

The second term is ≥ 0, with equality iff $\beta_1 = \cdots = \beta_p = 0$.

Proof. By Theorem 5.4(a) with $E\mathbf{y} = X\beta$, $\mathrm{Var}\,\mathbf{y} = \sigma^2 I$: $B = I - H$ has $\mathrm{tr}B = n - p - 1$, $X'BX = 0$, so $E(SSE) = (n - p - 1)\sigma^2$; $A = H - J_n/n$ has $\mathrm{tr}A = p$ and $X'AX = X'(I - J_n/n)X$ ($HX = X$). And $\beta'X'(I - J_n/n)X\beta = \|(I - J_n/n)X\beta\|^2$ vanishes iff all slopes vanish (centred columns independent). ∎

For $p = 1$, $\beta'X'(I - J_n/n)X\beta = \beta_1^2 S_{xx}$, recovering Chapter 5. The **coefficient of determination** is $R^2 = SSR/SST = 1 - SSE/SST$, and $\hat{\sigma}^2 = MSE$ is the **residual mean square**.

6.4 Properties of the estimators

Motivation. $\hat{\boldsymbol{\beta}}$ is linear in $\mathbf{y}$, so its moments are immediate; H shows how fits depend on observations.

Theorem 6.3

(a) $E\hat{\beta} = \beta$ and $\operatorname{Var}\hat{\beta} = \sigma^2(X'X)^{-1}$; writing $(X'X)^{-1} = (c_{ij})$, $\operatorname{Var}\hat{\beta}_j = c_{jj}\sigma^2$ and $\operatorname{Cov}(\hat{\beta}_i, \hat{\beta}_j) = c_{ij}\sigma^2$.

(b) At a point $\mathbf{x} = (1, x_1, \ldots, x_p)'$, $\hat{y} = \mathbf{x}'\hat{\beta}$ estimates $E(y \mid \mathbf{x})$ with $\operatorname{Var}\hat{y} = \mathbf{x}'(X'X)^{-1}\mathbf{x}\,\sigma^2$.

(c) A new observation y_s at $\mathbf{x}$ has $\operatorname{Var}(y_s - \hat{y}) = \{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}\sigma^2$.

Proof. (a) $E\hat{\beta} = (X'X)^{-1}X'X\beta = \beta$ and $\operatorname{Var}\hat{\beta} = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) $\operatorname{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}'\operatorname{Var}(\hat{\beta})\mathbf{x}$.

(c) y_s is independent of $\hat{y}$. ■

For $p = 1$: $\operatorname{Var}\hat{\beta}_0 = \sigma^2(1/n + \bar{x}^2/S_{xx})$, $\operatorname{Var}\hat{\beta}_1 = \sigma^2/S_{xx}$, $\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$, $\operatorname{Var}\hat{y} = \sigma^2\{1/n + (x - \bar{x})^2/S_{xx}\}$. The design enters through $(X'X)^{-1}$: spread x's and avoid near-collinear columns.

6.5 Distribution theory

Motivation. Under normal errors $\hat{\boldsymbol{\beta}}$ and SSE are independent with $SSE/\sigma^2 \sim \chi^2$, giving t and F tests.

Theorem 6.4

Under $\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$:

(a) $SSE/\sigma^2 \sim \chi^2(n - p - 1)$;

(b) $SSR/\sigma^2 \sim \chi^2(p, \lambda)$ with $\lambda = \beta'X'(I - J_n/n)X\beta/(2\sigma^2)$;

(c) $SST/\sigma^2 \sim \chi^2(n - 1, \lambda)$ with the same λ;

(d) SSR and SSE are independent, and $\hat{\beta}$ is independent of SSE;

(e) $F_0 = MSR/MSE \sim F(p, n{-}p{-}1, \lambda)$; under $H_0 : \beta_1 = \cdots = \beta_p = 0$, $F_0 \sim F(p, n{-}p{-}1)$.

Proof. All three matrices $I - H$, $H - J_n/n$ and $I - J_n/n$ are symmetric idempotent: for $H - J_n/n$ use $HJ_n = J_nH = J_n$ (since $H\mathbf{1} = \mathbf{1}$, the intercept column) and $J_n^2 = nJ_n$:

$$\Big(H - \frac{J_n}{n}\Big)^2 = H - \frac{J_n}{n} - \frac{J_n}{n} + \frac{J_n}{n} = H - \frac{J_n}{n}.$$

Their ranks (traces) are $n - p - 1$, p and $n - 1$. Theorem 5.5(b) gives (a)–(c), the noncentralities being $\frac{1}{2\sigma^2}(X\beta)'A(X\beta)$; for $I - H$ it is 0 because $(I - H)X = 0$.

(d) $(H - J_n/n)(I - H) = 0$, so Theorem 5.6(b) gives independence of SSR and SSE. Also $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ and $(X'X)^{-1}X'(I - H) = 0$, so Theorem 5.6(a) applies.

(e) follows from (a), (b), (d). ■

Consequently, with c_{jj} the diagonal of $(X'X)^{-1}$,

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{c_{jj}MSE}} \sim t(n - p - 1).$$

Example 6.2

In Example 6.1, $SST = \sum(y_i - 8.5)^2 = 73.5$ and $SSR = \hat\beta_1 S_{1y} + \hat\beta_2 S_{2y} = 2.167 \cdot 35.5 - 0.167 \cdot 28.5 = 72.17$, so $SSE = 1.33$ with 3 degrees of freedom and $R^2 = 0.982$. Then $F_0 = (72.17/2)/(1.33/3) = 81.2 > F_{0.05}(2,3) = 9.55$: the regression is highly significant.

6.6 Partial regression: the Frisch–Waugh–Lovell theorem

Partition $X = (X_1, X_2)$, with X_1 $(n \times (q+1))$ containing the intercept and X_2 $(n \times (p-q))$ the remaining columns, and write $\mathbf{y} = X_1\beta_1 + X_2\beta_2 + \epsilon$. Let $H_1 = X_1(X_1'X_1)^{-1}X_1'$.

Theorem 6.5 (Frisch–Waugh–Lovell)

Let $\mathbf{e} = (I - H_1)\mathbf{y}$ be the residuals of regressing $\mathbf{y}$ on X_1, and $X_{2\cdot1} = (I - H_1)X_2$ the residuals of regressing each column of X_2 on X_1. The least squares coefficient from regressing $\mathbf{e}$ on $X_{2\cdot1}$,

$$\tilde\beta_2 = (X_{2\cdot1}'X_{2\cdot1})^{-1}X_{2\cdot1}'\mathbf{e},$$

equals the $\hat\beta_2$ part of the full least squares estimator. Moreover the residuals of the two regressions coincide.

Proof [Supplement]. The normal equations of the full model are

$$\begin{pmatrix} X_2'X_2 & X_2'X_1 \\ X_1'X_2 & X_1'X_1 \end{pmatrix}\begin{pmatrix} \hat\beta_2 \\ \hat\beta_1 \end{pmatrix} = \begin{pmatrix} X_2'\mathbf{y} \\ X_1'\mathbf{y} \end{pmatrix}.$$

Multiply on the left by $\begin{pmatrix} I & -X_2'X_1(X_1'X_1)^{-1} \\ 0 & I \end{pmatrix}$. The first block row becomes

$$X_2'(I - H_1)X_2\hat\beta_2 + \{X_2'X_1 - X_2'H_1X_1\}\hat\beta_1 = X_2'(I - H_1)\mathbf{y},$$

and the bracket is 0 ($H_1X_1 = X_1$). So $\hat\beta_2 = \{X_2'(I-H_1)X_2\}^{-1}X_2'(I-H_1)\mathbf{y} = (X_{2\cdot1}'X_{2\cdot1})^{-1}X_{2\cdot1}'\mathbf{e} = \tilde\beta_2$. Residuals: applying $I - H_1$ to $\mathbf{y} - X_1\hat\beta_1 - X_2\hat\beta_2$ (orthogonal to X_1) gives $\mathbf{e} - X_{2\cdot1}\hat\beta_2$. ■

Interpretation

So $\hat\beta_j$ is the effect of x_j *after adjusting both y and x_j for the other regressors*: the slope of residuals on residuals. It equals the simple slope when x_j is orthogonal to the other centred regressors.

6.7 Selected exercises

Motivation. Independence of $\hat\beta$ and $\mathbf{e}$, R as the maximal correlation, and invariance to parametrization of the column space.

Exercise 6.1 (Joint distribution of $\hat\beta$ and e)

Under $\mathbf{y} \sim N_n(X\beta, \sigma^2 I_n)$, find the joint distribution of $(\hat\beta', \mathbf{e}')'$, with its mean vector and covariance matrix, and deduce that $\hat\beta$ and $\mathbf{e}$ are independent.

Solution.

Step 1. $\begin{pmatrix}\hat\beta\\ \mathbf{e}\end{pmatrix} = M\mathbf{y}$ with $M = \begin{pmatrix}(X'X)^{-1}X'\\ I-H\end{pmatrix}$, a fixed $(p+1+n)\times n$ matrix, so the vector is multivariate normal (Section 5.4).

Step 2. (Mean.) $E\hat\beta = \beta$ and $E\mathbf{e} = (I-H)X\beta = \mathbf{0}$, since $HX = X$.

Step 3. (Covariance.) $\mathrm{Var}(M\mathbf{y}) = \sigma^2 MM'$. The diagonal blocks are $(X'X)^{-1}X'X(X'X)^{-1} = (X'X)^{-1}$ and $(I-H)(I-H)' = I-H$; the off-diagonal block is $(X'X)^{-1}X'(I-H) = (X'X)^{-1}(X'-X'H) = 0$, because $X'H = (HX)' = X'$. Hence

$$\begin{pmatrix}\hat\beta\\ \mathbf{e}\end{pmatrix} \sim N_{p+1+n}\left(\begin{pmatrix}\beta\\ \mathbf{0}\end{pmatrix},\ \sigma^2\begin{pmatrix}(X'X)^{-1} & 0\\ 0 & I-H\end{pmatrix}\right).$$

The distribution is singular, since $I-H$ has rank $n-p-1 < n$.

Step 4. (Independence.) By the mgf characterization (3) of Section 5.4, for $\mathbf{t} = (\mathbf{t}_1', \mathbf{t}_2')'$ the joint mgf is

$$\exp\{\mathbf{t}_1'\beta + \tfrac{\sigma^2}{2}\mathbf{t}_1'(X'X)^{-1}\mathbf{t}_1\}\cdot\exp\{\tfrac{\sigma^2}{2}\mathbf{t}_2'(I-H)\mathbf{t}_2\},$$

so $\hat\beta \perp \mathbf{e}$, hence independent of SSE and residual plots (strengthening Theorem 6.4(d)). ■

Exercise 6.2 (Multiple correlation as a maximal correlation)

Let $Y_i = \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$ with centred data ($\sum_i Y_i = 0$, $\sum_i x_{ij} = 0$), X $n\times p$ of full rank, $n > p$; $\hat Y_i = \sum_j \hat\beta_j x_{ij}$, $SSR = \sum \hat Y_i^2$, $SST = \sum Y_i^2$, $\langle \mathbf{a}, \mathbf{b}\rangle = \sum a_i b_i$.
(a) Express SSR and SST through X and $\mathbf{Y}$.
(b) Assuming $SST \neq 0$, prove that

$$\max_{\beta\in\mathbb{R}^p,\ \beta\neq\mathbf{0}} \frac{\langle X\beta, \mathbf{Y}\rangle}{\|X\beta\|\,\|\mathbf{Y}\|} = \sqrt{\frac{SSR}{SST}}.$$

(c) Which β attain the maximum in (b)?

Solution.

(a) $\hat{\mathbf{Y}} = H\mathbf{Y}$ with $H = X(X'X)^{-1}X'$ symmetric idempotent, so

$$SSR = \mathbf{Y}'H'H\mathbf{Y} = \mathbf{Y}'X(X'X)^{-1}X'\mathbf{Y}, \qquad SST = \mathbf{Y}'\mathbf{Y}.$$

(b) For $\beta \neq \mathbf{0}$, $X\beta \neq \mathbf{0}$; as $HX\beta = X\beta$, symmetry of H and Cauchy–Schwarz give

$$\langle X\beta, \mathbf{Y}\rangle = \langle HX\beta, \mathbf{Y}\rangle = \langle X\beta, H\mathbf{Y}\rangle \le \|X\beta\|\,\|H\mathbf{Y}\| = \|X\beta\|\sqrt{SSR}.$$

so the ratio is at most $\sqrt{SSR/SST}$. If $SSR > 0$, $\beta = \hat\beta$ attains it; if $SSR = 0$, $H\mathbf{Y} = \mathbf{0}$ and the ratio is always 0.

(c) If $SSR > 0$, equality iff $X\beta = cX\hat\beta$, $c > 0$, i.e. $\beta = c\hat\beta$; if $SSR = 0$, every $\beta \neq \mathbf{0}$.

For centred data the ratio is the correlation of $\mathbf{Y}$ with $X\beta$: R is the largest correlation of y with a linear combination of regressors, attained by the fitted values. ■

Exercise 6.3 (Reparametrization does not change the fit)

Let A be a nonsingular $(p+1)\times(p+1)$ matrix and $Z = XA$.
(a) Show that $\hat\gamma = A^{-1}\hat\beta$ with $\mathrm{Var}\,\hat\gamma = \sigma^2 A^{-1}(X'X)^{-1}(A^{-1})'$, and that H, fitted values, residuals, SSE, MSE, R^2 are unchanged.

(b) With $z_j = (x_j - c_j)/d_j$ ($d_j \neq 0$) and the intercept kept, show that the new slopes are $d_j\hat{\beta}_j$ and the slope t statistics change at most in sign.

Solution [Supplement].

(a) Z has full column rank, because $Z\mathbf{a} = \mathbf{0}$ gives $XA\mathbf{a} = \mathbf{0}$, so $A\mathbf{a} = \mathbf{0}$ and $\mathbf{a} = \mathbf{0}$. Then

$$\hat{\gamma} = (A'X'XA)^{-1}A'X'\mathbf{y} = A^{-1}(X'X)^{-1}(A')^{-1}A'X'\mathbf{y} = A^{-1}\hat{\beta},$$

and $\operatorname{Var}\hat{\gamma} = \sigma^2(Z'Z)^{-1} = \sigma^2 A^{-1}(X'X)^{-1}(A^{-1})'$. The hat matrix is

$$Z(Z'Z)^{-1}Z' = XAA^{-1}(X'X)^{-1}(A')^{-1}A'X' = H,$$

so the fitted values $H\mathbf{y}$, the residuals $(I - H)\mathbf{y}$ and SSE coincide; MSE has the same degrees of freedom $n - p - 1$, and SST does not involve the design, so R^2 coincides too.

(b) The new design has columns $\mathbf{1}$ and $\mathbf{z}_j = (\mathbf{x}_j - c_j\mathbf{1})/d_j$, so $Z = XA$ with A upper triangular with diagonal $(1, 1/d_1, \ldots, 1/d_p)$, hence nonsingular. By (a) the fitted values agree:

$$\hat{\beta}_0\mathbf{1} + \sum_j \hat{\beta}_j\mathbf{x}_j = \hat{\gamma}_0\mathbf{1} + \sum_j \hat{\gamma}_j\frac{\mathbf{x}_j - c_j\mathbf{1}}{d_j}.$$

Matching coefficients (independent columns): $\hat{\gamma}_j = d_j\hat{\beta}_j$, $\hat{\gamma}_0 = \hat{\beta}_0 + \sum_j \hat{\beta}_j c_j$; likewise $\operatorname{Var}\hat{\gamma}_j = d_j^2 \operatorname{Var}\hat{\beta}_j$. With the same MSE,

$$\frac{\hat{\gamma}_j}{\sqrt{d_j^2 c_{jj} MSE}} = \frac{d_j}{|d_j|} \cdot \frac{\hat{\beta}_j}{\sqrt{c_{jj} MSE}}.$$

So centring and rescaling the regressors (as in Chapter 15) change the units of the slopes but no test or fitted value. ■